

Lecture 5 Tutorial — Counting Theories & Naive Bayes

Yaran Chen
Yaran.chen@xjtlu.edu.cn

Q1. Bayes' Rule Components (Matching)

Match each term to the correct name in the lecture.

Given:

$$P(c \mid x) = \frac{P(c)P(x \mid c)}{P(x)}$$

Terms:

- A. $P(c)$
- B. $P(x \mid c)$
- C. $P(c \mid x)$
- D. $P(x)$

Names:

1. Prior probability
2. Likelihood
3. Posterior probability
4. Evidence factor (observation)

Q1

$$P(c|x) = \frac{P(x|c) \times P(c)}{P(x)}$$

Posterior

$P(c|x)$

Probability of class 'c'
given data 'x'.
"Updated Belief"

Likelihood

$P(x|c)$

Probability of data 'x'
if class is 'c'.
"Pattern matched"

Prior

$P(c)$

Initial probability
of class 'c' overall.
"Initial Assumption"

Evidence

$P(x)$

Overall probability of
observing data 'x'.
"Normalization Factor"

Real-World Example: Spam Email Filter

Is the email Spam (c) given it contains the word "FREE" (x)?

1. **Prior $P(c)$** : Usually, 20% of all emails you receive are spam.
2. **Likelihood $P(x|c)$** : 80% of confirmed spam emails contain the word "FREE".
3. **Evidence $P(x)$** : Regardless of type, 25% of ALL emails contain the word "FREE".
4. **Posterior $P(c|x)$** : If you see an email with "FREE", what is the chance it is spam?

Calculation: $P(c|x) = (0.80 \times 0.20) / 0.25 = 0.64$ — (64% chance it is spam!)

Q2. Likelihood Computation (Coin Models)

An experiment result is:

$$D = \text{"THHTTTHTTH"}$$

Consider two candidate models:

$$M_1 : p(H) = 0.5, p(T) = 0.5$$

$$M_2 : p(H) = 0.4, p(T) = 0.6$$

1. Count the number of H and T in D .
2. Compute $L_1 = p(D \mid M_1)$.
3. Compute $L_2 = p(D \mid M_2)$.
4. Select the model with higher likelihood (circle one): M_1 / M_2 .

Likelihood Computation & Model Selection

Which coin model best explains the observed sequence?

Step 1: Observed Data (D)

T H H T T T H T T H

Count: H = 4 , T = 6

Step 2 & 3: Compute Likelihood $L = P(\text{Data} \mid \text{Model})$

Model 1 (Fair Coin)

$P(H) = 0.5$, $P(T) = 0.5$

$$L_1 = (0.5)^4 \times (0.5)^6$$

$$L_1 = (0.5)^{10} = 1 / 1024$$

$$L_1 \approx 0.000976$$

vs

Model 2 (Biased Coin)

$P(H) = 0.4$, $P(T) = 0.6$

$$L_2 = (0.4)^4 \times (0.6)^6$$

$$L_2 = 0.0256 \times 0.046656$$

$$L_2 \approx 0.001194$$

Conclusion: Model 2 is Selected

Because L_2 (0.001194) > L_1 (0.000976). The data is more likely under Model 2.

Q3. Log-likelihood (Numerical Stability)

Using the same D , compute the log-likelihood difference:

$$\Delta = \log p(D \mid M_2) - \log p(D \mid M_1)$$

1. Write Δ in terms of $\log(0.4)$, $\log(0.6)$, and $\log(0.5)$.
2. State whether $\Delta > 0$, $\Delta = 0$, or $\Delta < 0$ (no calculator required).
(Use any log base; the sign must be consistent.)

Log-Likelihood & Numerical Stability

Why do we take the Logarithm of probabilities in Machine Learning?

The Computer Underflow Problem ⚠

Multiplying many tiny probabilities causes computers to round to zero:

$$L = 0.5 \times 0.5 \times 0.5 \dots (100 \text{ times}) = 0.5^{100} \approx 0.0000000000\dots \text{ (Crashes!)}$$

Solution: $\log(A \times B) = \log(A) + \log(B)$. Multiplication becomes Addition!

Applying Logarithm to Our Coin Models (from Q2)

Model 1 (Fair Coin)

$$\log(L_1) = \log((0.5)^{10})$$

Law of powers: $\log(X^Y) = Y \cdot \log(X)$

$$\log(L_1) = 10 \log(0.5)$$

Model 2 (Biased Coin)

$$\log(L_2) = \log((0.4)^4 \times (0.6)^6)$$

Products to Sums: $\log(A \cdot B) = \log(A) + \log(B)$

$$\log(L_2) = 4 \log(0.4) + 6 \log(0.6)$$

Computing Delta (Δ)

$$\Delta = \log(L_2) - \log(L_1)$$

$$\Delta = 4 \log(0.4) + 6 \log(0.6) - 10 \log(0.5)$$

Because $L_2 > L_1$, and logarithm is strictly increasing...

Therefore: $\Delta > 0$

Q4. Maximum Likelihood Estimation (MLE) for a Coin

For Bernoulli trials, let $\theta = p(H)$. Using “THHTTTHTTH”:

1. Write the likelihood in the form $p(D \mid \theta) = \theta^k (1 - \theta)^{n-k}$.
2. Give the MLE $\hat{\theta}$ as a number.

(Answer only; no derivation required.)

Q4

Step 1: Write the Likelihood Function (H=4, T=6)

$$p(\text{Data} \mid \theta) = L(\theta) = \theta^4 \times (1 - \theta)^6 \quad (\text{Where } \theta \text{ is the unknown probability of getting Heads})$$

The estimated parameter ($\hat{\theta}$) is simply the empirical frequency:

$$\text{MLE } \hat{\theta} = k / n = 4 / 10 = 0.4$$

Q5. Occam's Razor / Size Principle (Model Likelihood)

Let

$$D = \{16, 8, 2, 64\}.$$

Consider two hypotheses:

h_{two} : “powers of two under 100”

h_{even} : “even numbers below 100”

Assume each hypothesis defines a finite set and is uniform over that set. That is, if x is consistent with h , then

$$p(x \mid h) = \frac{1}{|h|}.$$

1. Compute $|h_{\text{two}}|$.
2. Compute $|h_{\text{even}}|$.
3. Compute the likelihood ratio:

$$\frac{p(D \mid h_{\text{two}})}{p(D \mid h_{\text{even}})}.$$

(Leave your answer as an exact fraction or a power expression.)

Occam's Razor & The Size Principle

Why simpler models (smaller hypothesis spaces) are mathematically more probable.

Data D = {16, 8, 2, 64}



Uniform Distribution Likelihood

Probability of drawing exact N=4 items from size |h|:

$$P(D | h) = (1 / |h|)^N$$

$$P(D | h_{\text{even}}) = (1 / 49)^4$$

(It's extremely rare to pick exactly those 4 from 49 evens)

$$P(D | h_{\text{two}}) = (1 / 7)^4$$

(Very high chance to pick those 4 if the pool is only 7)

ratio

Comparing the two Likelihoods (Likelihood Ratio):

$$\text{Ratio} = (1 / 7)^4 \div (1 / 49)^4 = (49 / 7)^4 = 7^4 = 2401$$

Conclusion: The "Powers of 2" hypothesis is 2401 times more believable!

Q6. Posterior Update (Equal Priors)

Continue Q5. Assume equal priors:

$$p(h_{\text{two}}) = p(h_{\text{even}}) = 0.5$$

A new sample arrives:

$$x_{\text{new}} = 32$$

1. Compute the unnormalised posteriors:

$$p(h_{\text{two}} \mid D \cup \{32\}) \propto p(D \cup \{32\} \mid h_{\text{two}})p(h_{\text{two}})$$

$$p(h_{\text{even}} \mid D \cup \{32\}) \propto p(D \cup \{32\} \mid h_{\text{even}})p(h_{\text{even}})$$

2. Decide which posterior is larger (circle one): h_{two} / h_{even} .

Bayesian Posterior Update

Combining Initial Beliefs (Priors) with New Evidence (Likelihood)

Updated Data Set (D)

$D = \{ 16, 8, 2, 64 \} + \text{New Sample: } 32 \Rightarrow \text{Total } N = 5$

Bayes' Theorem: $\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$

Model 1: h_{two} (Powers of 2)

1. Prior: $P(h_{\text{two}})$ **0.5**

2. Likelihood: $P(D | h_{\text{two}})$ $(1/7)^5$

3. Posterior $\propto 0.5 \times (1/7)^5$
(Unnormalized)

vs

Model 2: h_{even} (Even Numbers)

1. Prior: $P(h_{\text{even}})$ **0.5**

2. Likelihood: $P(D | h_{\text{even}})$ $(1/49)^5$

3. Posterior $\propto 0.5 \times (1/49)^5$
(Unnormalized)

Conclusion: h_{two} has the LARGER posterior.

Because both Priors are equal (0.5), the Likelihood dictates the winner.

$(1/7)^5$ is massively larger than $(1/49)^5$

Q7. Entropy, Cross Entropy, KL Divergence (Fill in + Compute)

The lecture defines the KL divergence (relative entropy) between P and Q as:

$$KL(P\|Q) = \sum_{i=1}^{\ell} p_i \log \frac{p_i}{q_i}$$

Let

$$P = (0.5, 0.5), \quad Q = (0.9, 0.1).$$

1. Compute $KL(P\|Q)$ (use natural log or base-2 log; state the base).
2. State whether $KL(P\|Q)$ can ever be negative (Yes / No).

Kullback-Leibler (KL) Divergence

Measuring the "distance" (information loss) when using Q to approximate true distribution P.

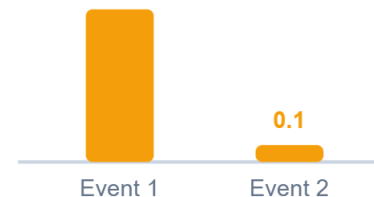
True Distribution (P)

$P = (0.5, 0.5)$



Approximation (Q)

$0.9Q = (0.9, 0.1)$



Step-by-Step Mathematical Computation

$$\begin{aligned} D_{KL}(P \parallel Q) &= P_1 \ln(P_1 / Q_1) + P_2 \ln(P_2 / Q_2) \\ &= 0.5 \ln(0.5 / 0.9) + 0.5 \ln(0.5 / 0.1) \\ &= 0.5 \ln(5 / 9) + 0.5 \ln(5) \\ &= 0.5 [\ln(5 / 9) + \ln(5)] \text{ (Log addition rule)} \\ &= 0.5 \ln(25 / 9) \text{ (Since } 5/9 * 5 = 25/9) \end{aligned} \quad = \ln(5 / 3)$$

Can KL Divergence be negative?

NO. By **Gibbs' Inequality**, $D_{KL}(P \parallel Q) \geq 0$.

It acts like a "distance" measuring information loss. Distance cannot be negative!

Q8. Zero-point Problem and Laplacian Correction (Short Calculation)

A discrete model has symbols $\ell = 3$ with alphabet $\{a, b, c\}$. In a training dataset, the counts for a class are:

$$N_a = 2, \quad N_b = 0, \quad N_c = 1$$

1. Compute the unsmoothed MLE probabilities $p(a)$, $p(b)$, $p(c)$.
2. Apply Laplacian correction (add-one smoothing) and compute the smoothed probabilities.

The Zero-point Problem & Laplacian Smoothing

How to prevent unseen events from destroying entire probability models.

1. Unsmoothed MLE (Maximum Likelihood Estimation)

Raw Counts: $N(a) = 2$, $N(b) = 0$, $N(c) = 1$ Total $N = 3$

$$P(a) = \frac{2}{3}$$

$$P(b) = \frac{0}{3} = 0$$

$$P(c) = \frac{1}{3}$$

⚠ THE "ZERO VETO": In Naive Bayes, multiplying by $P(b)=0$ will make the entire final probability exactly 0!

2. Laplacian Smoothing (Add-One)

$$P(\text{event}) = \frac{\text{Count} + 1}{\text{Total } N + |V|} \quad |V| = \text{Number of categories}$$

If we add 1 to every count:

$$\text{New } N(a) = 2 + 1 = 3$$

$$\text{New } N(b) = 0 + 1 = 1$$

$$\text{New } N(c) = 1 + 1 = 2$$

Notice the Total N:

We have 3 categories: $V = \{a, b, c\}$, so $|V| = 3$

$$\text{New Denominator} = \text{Total } N (3) + |V| (3) = 6$$

Smoothed Posteriors:

$$P(a) = \frac{3}{6} = 1/2$$

$$P(b) = \frac{1}{6}$$

$$P(c) = \frac{2}{6} = 1/3$$

Q9. MAP vs MLE (Multiple Choice)

Select the most accurate statement.

- A. MLE selects the model with the highest posterior probability.
- B. MAP selects the model with the highest likelihood only.
- C. MAP maximises $p(c \mid x)$ and uses a term proportional to $p(c)p(x \mid c)$.
- D. MAP ignores the prior probability.

MAP vs. MLE: The Tale of Two Estimators

How do we choose the best hypothesis 'c' given our data 'x'?

MLE

(Maximum Likelihood Estimation)

Maximize: $p(x | c)$

Focus:

"Which class 'c' makes this exact data 'x' most likely to appear?"

Data-Driven Only (Ignores Prior)

MAP

(Maximum A Posteriori)

Maximize: $p(c | x)$

Focus:

"Given this data 'x', which class 'c' is most probable in reality?"

Data + Prior Experience

The Mathematical Relationship (Bayes' Theorem)

$$p(c | x) \propto p(x | c) \times p(c)$$

MAP (Posterior) is proportional to MLE (Likelihood) times the Prior.

Q10. Naive Bayes Independence Assumption (Fill in)

Complete the Naive Bayes factorisation shown in the lecture:

If

$$x = (x_1, x_2, \dots, x_\ell),$$

then

$$p(x \mid c) = p(x_1 \mid c) p(x_2 \mid c) \cdots p(x_\ell \mid c).$$

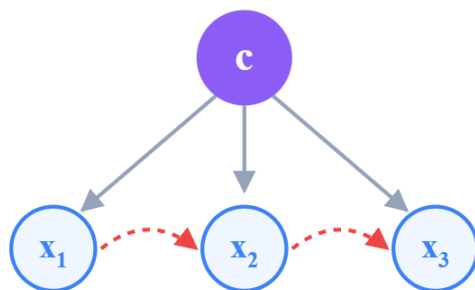
This factorisation is based on the assumption that the features are _____ given the class c .

The "Naive" Assumption of Naive Bayes

Features are conditionally independent given the class c .

Real World (Complex)

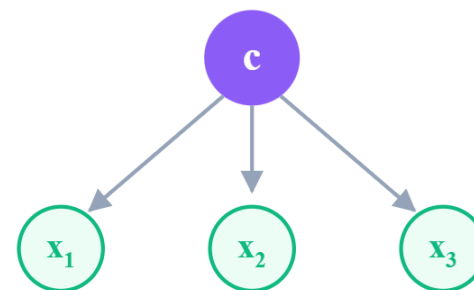
Features are often correlated



Messy Interdependencies

Naive Bayes (Simplified)

Features are fully independent



No Cross-Talk (Independent)

The Mathematical Transformation

Instead of calculating a massive, complex joint probability, we just multiply individual probabilities!

$$P(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3 | \mathbf{c})$$

becomes

$$P(\mathbf{x}_1 | \mathbf{c}) \times P(\mathbf{x}_2 | \mathbf{c}) \times P(\mathbf{x}_3 | \mathbf{c})$$

Q11. Naive Bayes Classification (Exam-style Computation)

Use the following training data (14 samples). Each row is:

(Outlook, Temperature, Humidity, Windy, Play)

Overcast	Hot	High	False	Yes
Overcast	Cool	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Rainy	Mild	Normal	False	Yes
Rainy	Mild	High	True	No
Sunny	Hot	High	True	No
Sunny	Hot	High	False	No
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Sunny	Mild	Normal	True	Yes

Classify the query:

$x = (\text{Outlook}=\text{Overcast}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{Normal}, \text{Windy}=\text{True})$

Part A (No smoothing)

1. Compute the unnormalised scores:

$$S_{\text{Yes}} = p(\text{Yes}) \prod_j p(x_j \mid \text{Yes})$$

$$S_{\text{No}} = p(\text{No}) \prod_j p(x_j \mid \text{No})$$

2. Which class is predicted by MAP? (Yes / No)

Part B (With Laplacian correction)

Apply add-one smoothing separately per feature within each class with category counts:

- Outlook: 3 categories (Sunny, Overcast, Rainy)
- Temperature: 3 categories (Hot, Mild, Cool)
- Humidity: 2 categories (High, Normal)
- Windy: 2 categories (True, False)

1. Recompute the two unnormalised scores S_{Yes} and S_{No} using the smoothed conditionals.
2. Which class is predicted now? (Yes / No)

Naive Bayes Classification: The smoothing effect

Computing raw probabilities vs. applying Laplacian Correction (+1)

Target Query (x):

Outlook = Overcast, Temp = Cool, Humidity = Normal, Windy = True

Part A: No Smoothing

Score S_{Yes} :

$= P(Yes) \times P(Overcast|Y) \times P(Cool|Y) \dots$

$= (9/14) \times (4/9) \times (3/9) \times (6/9) \times (3/9)$

≈ 0.02116

Score S_{No} :

$= P(No) \times P(Overcast|N) \times P(Cool|N) \dots$

$= (5/14) \times (0/5) \times (1/5) \times (1/5) \times (3/5)$

$= 0$

⚠ The "Zero Veto" wiped out the probability.

Part B: Laplacian Smoothing

Score S_{Yes} :

Conditionals $= (Count + 1) / (Total\ N + |V|)$

$= (9/14) \times (5/12) \times (4/12) \times (7/11) \times (4/11)$

≈ 0.02066

Score S_{No} :

Rescuing the 0 count using add-one:

$= (5/14) \times (1/8) \times (2/8) \times (2/7) \times (4/7)$

≈ 0.00182

✅ The absolute zero is fixed to a realistic 1/8.

Final MAP Predictions

Part A Winner:

YES ($0.02116 > 0$)

Part B Winner:

YES ($0.02066 > 0.00182$)

Q12. Bayesian Network (Multiple Choice)

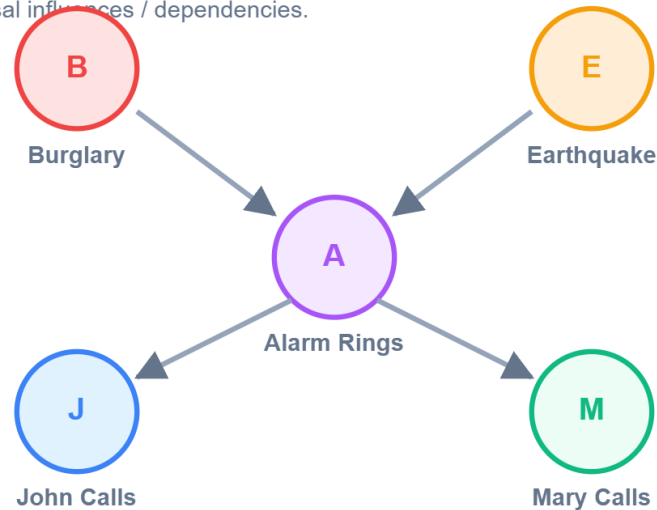
A Bayesian Network is described as using a Directed Acyclic Graph (DAG) to represent dependencies and factorise a joint distribution.

Which statement is most consistent with this description?

- A. It replaces probability with deterministic rules.
- B. It encodes feature dependencies via a DAG and expresses a joint distribution as a product of conditional distributions.
- C. It assumes all features are independent without exception.
- D. It can only be used for coin and dice experiments.

1. The Graph (DAG Structure)

Arrows represent direct causal influences / dependencies.



DAG Rules:

- ✓ Has Directions
- ✗ No Closed Loops

2. Expressing the Joint Distribution

Instead of calculating a massive table of all possible combinations $P(B, E, A, J, M)$, the Bayes Net states:

"The probability of any node only depends on its direct Parents."

$$P(B, E, A, J, M) = P(B) \times P(E) \times P(A | B, E) \times P(J | A) \times P(M | A)$$

(Notice how 'A' acts as a bottleneck. J and M only care about A, not B or E!)

Solution

B. It encodes feature dependencies via a DAG and expresses a joint distribution as a product of conditional distributions.